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ASSIGNMENT 3

Problem Statement: Write a program in C to generate multiplication tables for a given no.

Objectives:
- To understand the use of looping statements in C
- To generate multiplication tables using loops.
- To improve program logic & execution flow understanding.

Theory:

Syntax for for Loop example program.

~~Syntax for while loop example program~~

```
for (initialization; condition; increment/decrement)
  {
    statements;
  }
```

Syntax for while loop example program

```
initialization;
while (condition)
  {
    statements;
    increment/decrement;
  }
```

Syntax for do while loop example program

```
initialization;
do
{   statements;
    increment/decrement;
}
```

while (condition);

Algorithm

Step1: START
Step2: Declare variables n & i
Step3: Read the number n.
Step4: Initialize i=1
Step5: Repeat while i is less than equal to 10
- print n x i
- Increment of i by 1

Step 6: STOP.

C program:

```
# include <stdio.h>
int main() {
    int n, i;
    printf("Enter a num:");
    scanf("%d", &n);
    printf("Multiplication Table of %d:\n", n);
    for (i=1; i<=10; i++)
    { printf("%d x %d = %d\n", n, i, n*i);
    }
    return 0;
}
```

Platform:
Linux

Input:
Enter a num:

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Output: Displays the multiplication table of given number from 1 to 10.

Sample Input: 5

Sample Output:
Multiplication Table of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Conclusion: Thus, the C program successfully generates the multiplication table of a given number using looping statements.

FAQs
Q. Which loop is best suited for generating multiplication tables?
A. The for loop is best suited for generating multiplication tables because the number of iterations is known in advance & initialization, condition, and increment are written in a single line.

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2 What is the role of loop initialization?

A a) Loop initialization is used to set the starting value of the loop control variable before the loop begins execution.

b) It ensures that the loop starts from a known & correct state.

3) Difference between for loop, while loop & do while loop?

A.

For loop	While loop.
a) Entry-controlled loop	a) Entry-controlled loop.
b) Used when iterations are known	b) Used when iterations unknown.
c) Initialization, condition, increment in one line.	c) Initialization done separately.

4) Can a do-while loop be used to generate multiplication tables? Explain.

A. Yes, a do-while loop can be used to generate multiplication tables because it executes the loop body atleast once before checking the condition, making it suitable for table generation.

5) Why is the do-while loop called an exit-controlled loop?

A. The do-while loop is called an exit-controlled loop because the condition is checked after executing the loop body, ensuring the loop runs at once.